DBMS: Key Definitions

Database Management Systems basics

1. Database

A database is an organised collection of related data stored electronically so it can be easily accessed, managed and updated. Example: a college database holding students, courses and marks.

2. DBMS (Database Management System)

A DBMS is software that lets users create, store, retrieve, update and control data in a database. It acts as an interface between the user or application and the stored data. Examples: MySQL, PostgreSQL, Oracle, SQLite.

3. Schema

A schema is the logical structure or blueprint of a database. It defines the tables, columns, data types, relationships and constraints, but not the data itself. The schema rarely changes.

4. Data Independence

Data independence is the ability to change the schema at one level of the database without having to change the schema at the next higher level. It comes in two forms: physical (storage changes don't affect the logical schema) and logical (logical changes don't affect application programs).

Extra Content

Goes beyond the definitions on page 1

1. Three-Level Architecture

- **External level (view):** what each user or application sees; many views can exist for one database.
- **Conceptual level (logical):** the whole database structure, entities, relationships and constraints.
- **Internal level (physical):** how data is stored on disk, including files, indexes and access paths.

2. Database Languages

Language	Purpose	Commands
DDL	Define and change structure	CREATE, ALTER, DROP, TRUNCATE
DML	Read and modify data	SELECT, INSERT, UPDATE, DELETE
DCL	Control access and permissions	GRANT, REVOKE
TCL	Manage transactions	COMMIT, ROLLBACK, SAVEPOINT

3. Keys in the Relational Model

- **Super key:** any set of attributes that uniquely identifies a tuple.
- **Candidate key:** a minimal super key, with no redundant attribute.
- **Primary key:** the candidate key chosen to identify rows; it cannot be NULL.
- **Foreign key:** refers to the primary key of another table, enforcing referential integrity.

4. Normalization

Organising tables to reduce redundancy and avoid insertion, update and deletion anomalies.

- **1NF:** atomic values only; no repeating groups.
- **2NF:** in 1NF, with no partial dependency on part of a composite key.
- **3NF:** in 2NF, with no transitive dependency between non-key attributes.
- **BCNF:** for every functional dependency $X \rightarrow Y$, X must be a super key.

5. Transactions and ACID Properties

- **Atomicity:** all operations of a transaction succeed, or none do.
- **Consistency:** the database moves from one valid state to another.
- **Isolation:** concurrent transactions do not interfere with each other.
- **Durability:** once committed, changes survive crashes.